

OLLSCOIL NA hÉIREANN, CORCAIGH
THE NATIONAL UNIVERSITY OF IRELAND, CORK

COLÁISTE NA hOLLSCOILE, CORCAIGH
UNIVERSITY COLLEGE, CORK

SEAT NUMBER

Examination Session and Year	Autumn 2025							
Module Code	EC4215							
Module Title	Business Econometrics and Forecasting							
Paper Number	1							
External Examiner	Prof. Mac an Bhaird							
Head of School/ Department	Prof. Doran							
Internal Examiners	Dr. Turner							
Instructions to Candidates	<table border="1" style="width: 100%; border-collapse: collapse; margin-bottom: 5px;"> <tr> <th style="width: 20%;">Sections</th><th style="width: 40%;">No. of Questions to be attempted</th><th style="width: 40%;">% of total marks available</th></tr> <tr> <td style="text-align: center;">A</td><td>Long Questions 2 out of 3</td><td style="text-align: center;">100%</td></tr> </table> All questions within each section carry equal weight Attempt the correct number of questions in each section: Only that number of questions will be graded. If you attempt more than the correct number, please indicate what questions you wish to have graded. If you fail to do this, the appropriate number of <u>lowest-numbered questions</u> of those you answered will be graded.		Sections	No. of Questions to be attempted	% of total marks available	A	Long Questions 2 out of 3	100%
Sections	No. of Questions to be attempted	% of total marks available						
A	Long Questions 2 out of 3	100%						
Duration of Paper	1.5 hours							
Special Requirements	Non-programmable calculators may be used							
Question Paper to be handed up	YES <input style="width: 30px; border: 1px solid black;" type="checkbox"/> NO <input checked="" style="width: 30px; border: 1px solid black;" type="checkbox"/>							

**PLEASE DO NOT TURN THIS PAGE UNTIL INSTRUCTED TO DO SO
THEN ENSURE THAT YOU HAVE THE CORRECT EXAM PAPER**

Section A: Long Questions (Answer 2 out of 3)

- A1. (a)** Suppose that a researcher is examining the demand for flights from Ireland to Spain over time and she produces the following econometric equation (standard errors in brackets):

$$F_t = 150.8 + 5.2GDP_t + 11.2R_{t-1} - 10.4P_t$$

(2.4) (4.2) (6.8)

$$\bar{R}^2 = 0.67 \quad n = 20$$

Where:

F_t = the number of seats booked on flights from Ireland to Spain in year t (in thousands);

GDP_t = the value of Gross Domestic Product in Ireland in year t (in €bn);

R_{t-1} = the annual rainfall in Ireland in year $t-1$ (in millimetres); and

P_t = the average price of an adult seat on flights from Ireland to Spain in year t (in €).

- (i) Carefully interpret the above findings. **(35 per cent)**
- (ii) Does the above equation fit your expectations? Why or why not? **(15 per cent)**
- (iii) If you could add one more variable into the above regression, what would it be and why? **(10 per cent)**

- (b)** Suppose that a researcher runs a regression to examine the factors that affect weekly cinema attendance in Ireland over the course of a year, and gets the following results (standard errors in brackets):

$$A_t = 996.1 - 4.78P_t + 1.47S_t + 3.45H_t$$

(2.22) (0.84) (1.14)

$$\bar{R}^2 = 0.478 \quad n = 52$$
$$RSS = 49,016 \quad ESS = 45,443$$

Where

A_t = Sales of cinema tickets in week t (000s)

P_t = The average price of a cinema ticket in week t (€)

S_t = The average price of a streaming service in week t (€)

H_t = A dummy variable set to 1 if the week falls during the school holidays, and 0 otherwise.

Question A1 (b) continued on next page

Question A1 (b) continued

- (i) Looking at this output (without carrying out any statistical tests), would you expect the regression to be significant? Explain your answer.

(10 per cent)

- (ii) Formally test this regression for significance at the 5% level of significance. Explain the steps you take.

(30 per cent)

- A2. (a)** Suppose a researcher has run a regression to examine the factors affecting the monthly demand for electric cars in Ireland over time and gets the following results (standard errors in brackets):

$$EC_t = 80.4 - 4.4 PC_t - 2.1 PE_t + 15.8 EM_t \quad (Eqn. 2.1)$$

(1.8) (1.3) (5.4)

$N = 24$ Adjusted $R^2 = 0.551$

Where:

EC_t = the number of electric cars (000s) sold in month t

PC_t = the average price of electric cars in month t (€000)

PE_t = the average price of electricity (cent per kWh) in month t

EM_t = the number of people in employment in month t

Suppose the researcher is debating whether to include a variable to measure the public car charging infrastructure. If the researcher adds in this variable, she gets the following results:

$$EC_t = 67.1 - 4.5 PR_t - 1.9 PE_t + 15.7 EM_t + 2.1 CP_t \quad (Eqn. 2.2)$$

(1.7) (0.8) (5.2) (0.8)

$N = 24$ Adjusted $R^2 = 0.601$

Where:

CP_t = the number of public car charges available (in hundreds) in month t

All of the other variables are as before.

Should the researcher include CP_t in her regression (i.e. go with Eqn. 2.2 instead of Eqn. 2.1)? Explain your answer.

(70 per cent)

Question A2 continued on next page

Question A2 continued

- (b) A researcher was examining the factors affecting new car sales in Cork city and estimated the following regression:

$$\text{SALES}_t = 325.4 + 21.4P_t - 13.1 PF_t$$

Where:

SALES_t = the number of new car sales in Cork city in year t

P_t = the average price of a new car for sale in Cork city in year t

PF_t = the average price per litre of fuel (petrol and diesel) in Cork city in year t

She is surprised at the positive coefficient on the price of new cars, so she thinks she might have omitted a relevant variable. Two possible variables she thinks she might have omitted are PUB_t and INC_t , where:

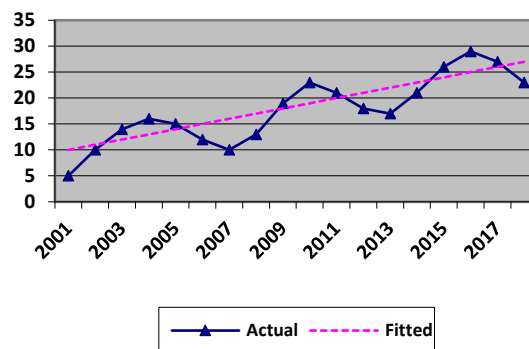
PUB_t = the number of public transport journeys taken in Cork city in year t

INC_t = the average household income in Cork city in year t .

Use expected bias analysis to determine whether either of these is a viable option as an omitted relevant variable.

(30 per cent)

- A3. (a) Suppose that a researcher runs a regression on a time-series dataset from 2001 to 2018 and gets the following actual and fitted values:



Question A3(a) continued on next page

Question A3(a) continued

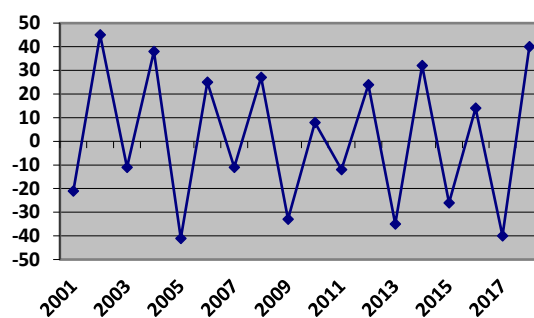
- (i) Is this regression likely to suffer from any econometric problem, and if so, which one?

(15 per cent)

- (ii) What is the likely cause of this econometric problem in this case? Explain your reasoning.

(15 per cent)

- (b) Below is a graph of residuals from a time-series regression.



- (i) What does this graph of residuals suggest? Explain your answer.

(15 per cent)

- (ii) Explain, with the aid of an example, why this might occur.

(15 per cent)

- (c) For each of the following scenarios, state whether autocorrelation is present, based on the Durbin Watson test with a 5% level of significance. Briefly explain your answer.

- (i) $d = 1.27$, $n = 20$, $k = 3$
- (ii) $d = 3.24$, $n = 30$, $k = 4$
- (iii) $d = 2.22$, $n = 22$, $k = 2$
- (iv) $d = 1.87$, $n = 35$, $k = 3$

(20 per cent)

Question A3 continued on next page

Question A3 continued

- (d) Suppose a researcher wants to examine finishing positions of a club in a professional soccer league over time, and collects data on a number of variables, proposing the following model:

$$FP_t = \beta_0 + \beta_1 FP_{t-1} + \beta_2 TS_t + \beta_3 MS_t + \beta_4 MA_t + \varepsilon_t$$

Where:

FP_t = the finishing position of the club in the league in year t

TS_t = the net transfer spending (in €m) of the club in year t

MS_t = the merchandise sales (in €m) of the club in year t

MA_t = the average weekly match attendances (in 000s) in year t

ε_t = a stochastic error term

Given that this is a time series regression, the researcher believes it may suffer from autocorrelation and wants to test this formally. How would you recommend the researcher goes about this in this case? Justify your answer.

(20 per cent)

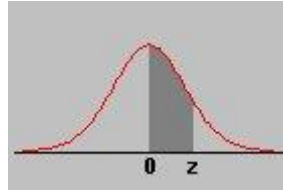
Appendix

A1: Selected Formulae

Slope coefficient: $\hat{\beta}_1 = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2}$	Intercept coefficient: $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$
Standard Error of slope coefficient: $se(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum x_i^2}}$	Standard Error of intercept coefficient: $se(\hat{\beta}_0) = \sqrt{\frac{\sum X_i^2}{n \sum x_i^2}} \hat{\sigma}$
Standard Error of estimated regression: $\hat{\sigma} = \sqrt{\frac{\sum e_i^2}{n - k - 1}}$	Total, Explained and Residual Sum of Squares: $\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n e_i^2$
<u>Confidence Interval:</u> Confidence Interval = $\hat{\beta}_k \pm t_{crit} * se(\hat{\beta}_k)$	<u>t-test:</u> $t = \frac{\hat{\beta}_k - \beta_{H0}}{se(\hat{\beta}_k)} \sim t(n - k - 1) df$
<u>F test (overall significance):</u> $F = \frac{ESS/k}{RSS/(n - k - 1)} \sim F(k, n - k - 1)$	<u>F test (overall significance):</u> $F = \frac{R^2/(k)}{(1 - R^2)/(n - k - 1)} \sim F(k, n - k - 1)$
<u>F test (Linear Restrictions):</u> $F = \frac{(RSS_R - RSS_{UR})/m}{RSS_{UR}/(n - k - 1)} \sim F(m, n - k - 1)$	<u>F test (Linear Restrictions):</u> $F = \frac{(R_{UR}^2 - R_R^2)/m}{(1 - R_{UR}^2)/(n - k - 1)} \sim F(m, n - k - 1)$

A2: Areas Under Normal Curve

Area between 0 and z

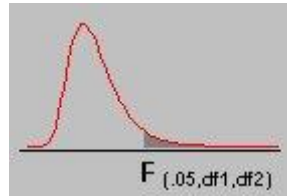


Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

A3: Student t- Distribution

d.f.	Level of Significance for One-Tailed Test						
	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
	Level of Significance for Two-Tailed Test						
	0.2	0.1	0.05	0.02	0.01	0.002	0.001
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.920	4.303	6.965	9.925	22.328	31.600
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.689
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.660
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
32	1.309	1.694	2.037	2.449	2.738	3.365	3.622
34	1.307	1.691	2.032	2.441	2.728	3.348	3.601
36	1.306	1.688	2.028	2.434	2.719	3.333	3.582
38	1.304	1.686	2.024	2.429	2.712	3.319	3.566
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
50	1.299	1.676	2.009	2.403	2.678	3.261	3.496
75	1.293	1.665	1.992	2.377	2.643	3.202	3.425
100	1.290	1.660	1.984	2.364	2.626	3.174	3.390
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.290

A4: Critical Values of the F Distribution at a 5 Percent Level of Significance



	Degrees of Freedom for the Numerator																			
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	inf	
Degrees of Freedom for the Denominator	1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
	2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
	3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
	26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
	27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
	28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
	29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
	120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
	Inf	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

A5: Durbin-Watson d Statistic: Significance Points of dL and dU at 0.05 Level of Significance

k = numbers of parameters estimated in the model

n	k = 2		k = 3		k = 4		k = 5		k = 6		k = 7	
	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU
6	0.610	1.400										
7	0.700	1.356										
8	0.763	1.332	0.467	1.896								
9	0.824	1.320	0.559	1.777	0.367	2.287						
10	0.879	1.320	0.629	1.699	0.455	2.128	0.296	2.588				
11	0.927	1.320	0.697	1.641	0.525	2.016	0.376	2.414	0.243	2.822		
12	0.971	1.324	0.758	1.604	0.595	1.928	0.444	2.283	0.315	2.645	0.203	3.004
13	0.971	1.331	0.812	1.579	0.658	1.864	0.512	2.177	0.380	2.506	0.268	2.832
14	1.010	1.340	0.861	1.562	0.715	1.816	0.574	2.094	0.444	2.390	0.328	2.692
15	1.045	1.350	0.905	1.551	0.767	1.779	0.632	2.030	0.505	2.296	0.389	2.572
16	1.077	1.361	0.946	1.543	0.814	1.750	0.685	1.977	0.562	2.220	0.447	2.471
17	1.106	1.371	0.982	1.539	0.857	1.728	0.734	1.935	0.615	2.157	0.502	2.388
18	1.133	1.381	1.015	1.536	0.897	1.710	0.779	1.900	0.664	2.104	0.554	2.318
19	1.158	1.391	1.046	1.535	0.933	1.696	0.820	1.872	0.710	2.060	0.603	2.258
20	1.180	1.401	1.074	1.536	0.967	1.685	0.859	1.848	0.752	2.023	0.649	2.206
21	1.201	1.411	1.100	1.537	0.998	1.676	0.894	1.828	0.792	1.991	0.691	2.162
22	1.221	1.420	1.125	1.538	1.026	1.669	0.927	1.812	0.829	1.964	0.731	2.124
23	1.239	1.429	1.147	1.541	1.053	1.664	0.958	1.797	0.863	1.940	0.769	2.090
24	1.257	1.437	1.168	1.543	1.078	1.660	0.986	1.785	0.895	1.920	0.804	2.061
25	1.273	1.446	1.188	1.546	1.101	1.656	1.013	1.775	0.925	1.902	0.837	2.035
26	1.288	1.454	1.206	1.550	1.123	1.654	1.038	1.767	0.953	1.886	0.868	2.013
27	1.302	1.461	1.224	1.553	1.143	1.652	1.062	1.759	0.979	1.873	0.897	1.992
28	1.316	1.469	1.240	1.556	1.162	1.651	1.084	1.753	1.004	1.861	0.925	1.974
29	1.328	1.476	1.255	1.560	1.181	1.650	1.104	1.747	1.028	1.850	0.951	1.959
30	1.341	1.483	1.270	1.563	1.198	1.650	1.124	1.743	1.050	1.841	0.975	1.944
31	1.352	1.489	1.284	1.567	1.214	1.650	1.143	1.739	1.071	1.833	0.998	1.931
32	1.363	1.496	1.297	1.570	1.229	1.650	1.160	1.735	1.090	1.825	1.020	1.920
33	1.373	1.502	1.309	1.574	1.244	1.650	1.177	1.732	1.109	1.819	1.041	1.909
34	1.383	1.508	1.321	1.577	1.258	1.651	1.193	1.730	1.127	1.813	1.061	1.900
35	1.393	1.514	1.333	1.580	1.271	1.652	1.208	1.728	1.144	1.808	1.079	1.891
36	1.402	1.519	1.343	1.584	1.283	1.653	1.222	1.726	1.160	1.803	1.097	1.884
37	1.411	1.525	1.354	1.587	1.295	1.654	1.236	1.724	1.175	1.799	1.114	1.876
38	1.419	1.530	1.364	1.590	1.307	1.655	1.249	1.723	1.190	1.795	1.131	1.870
39	1.427	1.535	1.373	1.594	1.318	1.656	1.261	1.722	1.204	1.792	1.146	1.864
40	1.435	1.540	1.382	1.597	1.328	1.658	1.273	1.722	1.218	1.789	1.161	1.859
41	1.442	1.544	1.391	1.600	1.338	1.659	1.285	1.721	1.230	1.786	1.175	1.854
42	1.449	1.549	1.399	1.603	1.348	1.660	1.296	1.720	1.243	1.784	1.189	1.849
43	1.456	1.553	1.407	1.606	1.357	1.662	1.306	1.720	1.255	1.781	1.202	1.845
44	1.463	1.558	1.415	1.609	1.366	1.663	1.317	1.720	1.266	1.779	1.215	1.841
45	1.469	1.562	1.423	1.612	1.375	1.665	1.326	1.720	1.277	1.778	1.227	1.838
46	1.475	1.566	1.430	1.615	1.383	1.666	1.336	1.720	1.287	1.776	1.238	1.835
47	1.481	1.570	1.437	1.618	1.391	1.668	1.345	1.720	1.298	1.775	1.250	1.832
48	1.487	1.574	1.444	1.620	1.399	1.669	1.354	1.720	1.307	1.774	1.260	1.829
49	1.493	1.578	1.450	1.623	1.406	1.671	1.362	1.721	1.317	1.773	1.271	1.826
50	1.498	1.581	1.456	1.626	1.414	1.672	1.370	1.721	1.326	1.772	1.281	1.824
51	1.503	1.585	1.462	1.628	1.421	1.674	1.378	1.721	1.335	1.771	1.291	1.822
52	1.509	1.588	1.468	1.631	1.427	1.675	1.386	1.722	1.343	1.770	1.300	1.820
53	1.514	1.592	1.474	1.633	1.434	1.677	1.393	1.722	1.351	1.769	1.309	1.818
54	1.518	1.595	1.480	1.636	1.440	1.678	1.400	1.723	1.359	1.769	1.318	1.817
55	1.523	1.598	1.485	1.638	1.446	1.680	1.407	1.723	1.367	1.768	1.326	1.815
56	1.528	1.601	1.490	1.641	1.452	1.681	1.414	1.724	1.374	1.768	1.334	1.814
57	1.532	1.605	1.495	1.643	1.458	1.683	1.420	1.725	1.382	1.768	1.342	1.812
58	1.536	1.608	1.500	1.645	1.464	1.684	1.426	1.725	1.389	1.768	1.350	1.811
59	1.540	1.610	1.505	1.647	1.469	1.686	1.433	1.726	1.395	1.767	1.358	1.810
60	1.545	1.613	1.510	1.650	1.474	1.687	1.438	1.727	1.402	1.767	1.365	1.809
61	1.549	1.616	1.514	1.652	1.480	1.689	1.444	1.727	1.408	1.767	1.372	1.808
62	1.552	1.619	1.519	1.654	1.485	1.690	1.450	1.728	1.415	1.767	1.379	1.807
63	1.556	1.622	1.523	1.656	1.490	1.692	1.455	1.729	1.421	1.767	1.385	1.807
64	1.560	1.624	1.527	1.658	1.494	1.693	1.461	1.730	1.427	1.767	1.392	1.806
65	1.563	1.627	1.532	1.660	1.499	1.695	1.466	1.730	1.432	1.767	1.398	1.805
66	1.567	1.629	1.536	1.662	1.503	1.696	1.471	1.731	1.438	1.767	1.404	1.805
67	1.570	1.632	1.539	1.664	1.508	1.697	1.476	1.732	1.443	1.767	1.410	1.804
68	1.574	1.634	1.543	1.666	1.512	1.699	1.481	1.733	1.449	1.768	1.416	1.804
69	1.577	1.637	1.547	1.668	1.516	1.700	1.485	1.733	1.454	1.768	1.422	1.803
70	1.580	1.639	1.551	1.670	1.521	1.701	1.490	1.734	1.459	1.768	1.427	1.803
75	1.583	1.641	1.554	1.672	1.525	1.703	1.494	1.735	1.464	1.768	1.433	1.802
80	1.598	1.652	1.571	1.680	1.543	1.709	1.515	1.739	1.487	1.770	1.458	1.801
85	1.611	1.662	1.586	1.688	1.560	1.715	1.534	1.743	1.507	1.772	1.480	1.801
90	1.624	1.671	1.600	1.696	1.575	1.721	1.550	1.747	1.525	1.774	1.500	1.801
95	1.635	1.679	1.612	1.703	1.589	1.726	1.566	1.751	1.542	1.776	1.518	1.801
100	1.645	1.687	1.623	1.709	1.602	1.732	1.579	1.755	1.557	1.778	1.535	1.802
150	1.654	1.694	1.634	1.715	1.613	1.736	1.592	1.758	1.571	1.780	1.550	1.803
200	1.758	1.778	1.748	1.789	1.738	1.799	1.728	1.810	1.718	1.820	1.707	1.831

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